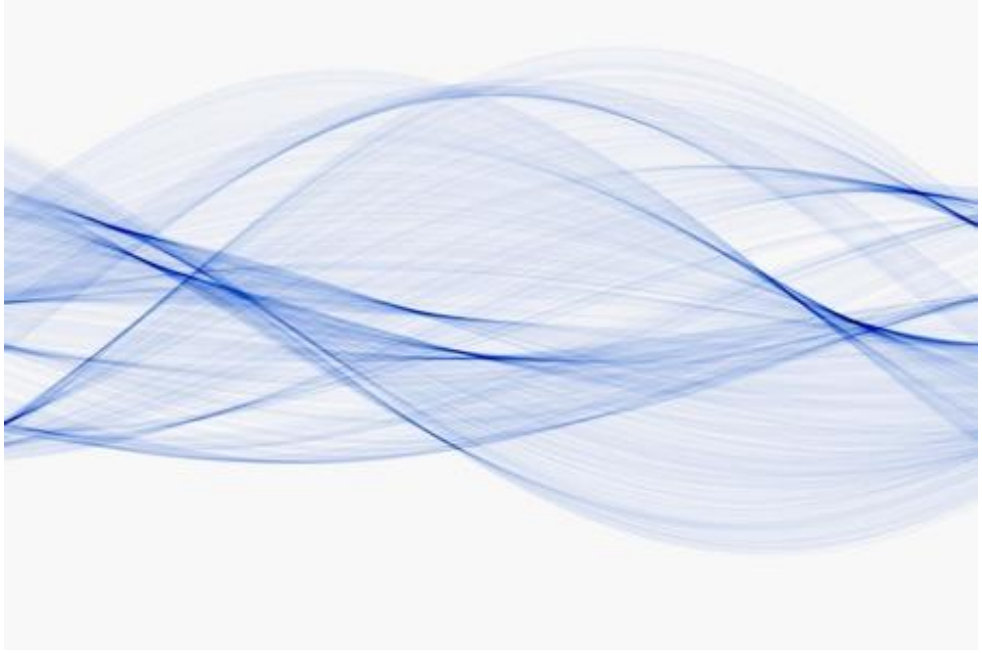
Fiber Materials and Fabrication Method

Dr. Monir Morshed
Professor
Dept. of ICT, MBSTU



Fiber Materials

Requirements of Fiber Optic Material:

i. The material must be transparent for efficient transmission of light.

ii. It must be possible to draw long thin fibers from the material.

iii. Fiber material must be compatible with the cladding material.

Most useful fiber materials are Glass and plastics. They fulfill the above requirements, however- Various types of high loss and low loss glass fibers are available to suit the requirements and Plastic fibers are not popular because of high attenuation they have better mechanical strength

Glass Fibers:

- Glass is made by fusing mixtures of metal oxides having refractive index of 1.458 at 850 nm.
- For changing the refractive index different oxides such as B_2O_3, GeO_2 and P_2O_5 are added as dopants.

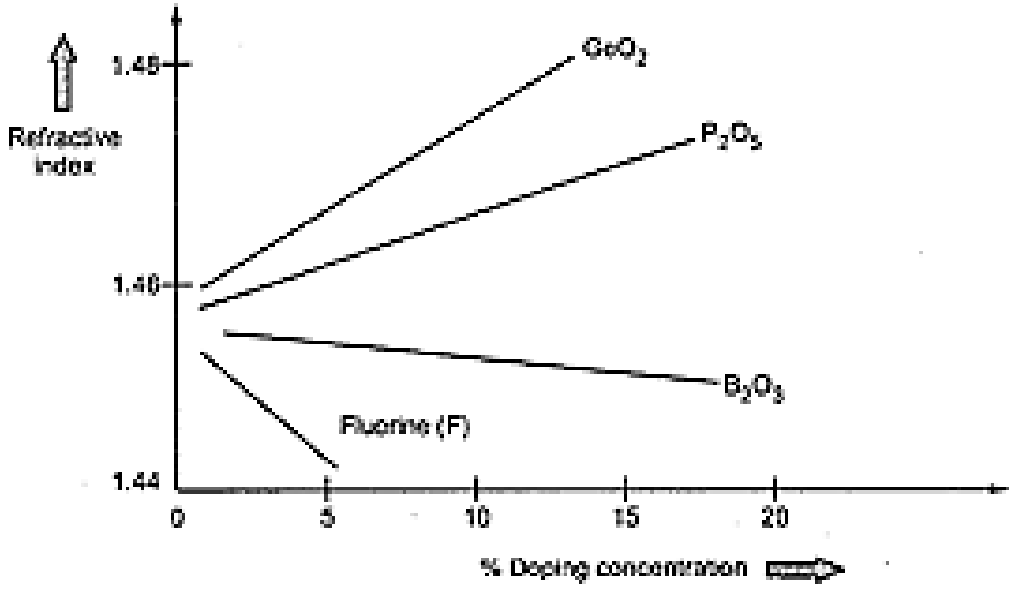


Fig. 20 Variation of refractive index with doping concentration

Fiber Materials

Compositions:

Composition	Core	Cladding
1	GeO_2-SiO_2	SiO_2
2	P_2O_5-SiO_2	SiO_2
3	SiO_2	B_2O_3-SiO_2
4	GeO_2-B_2O_3-SiO_2	B_2O_3-SiO_2

The principal raw material for silica is sand and glass. The fiber composed of pure silica is called as silica glass. The desirable properties of silica glass are :-

- Resistance to deformation even at high temperature.
- Resistance to breakage from thermal shocks (low thermal expansion).
- Good chemical durability.
- Better transparency.

Other types of glass fibers are :

- Halide glass fibers
- Active glass fibers
- Chalgenide glass fibers
- Plastic optical fibers

Fiber Fabrication Method

There are three steps of fiber fabrication:

- Fabrication of the preform
- Drawing the fiber from the preform
- Coating and jacketing process

Fabrication of the preform: The process where vapours of metal halides such as $SiCl_4$ and $Gecl_4$ reactive with oxygen and forms powder of SiO_2 particles. The SiO_2 particles are collected on surface of bulk glass and then sintered to form a glass rod called **Preform.**

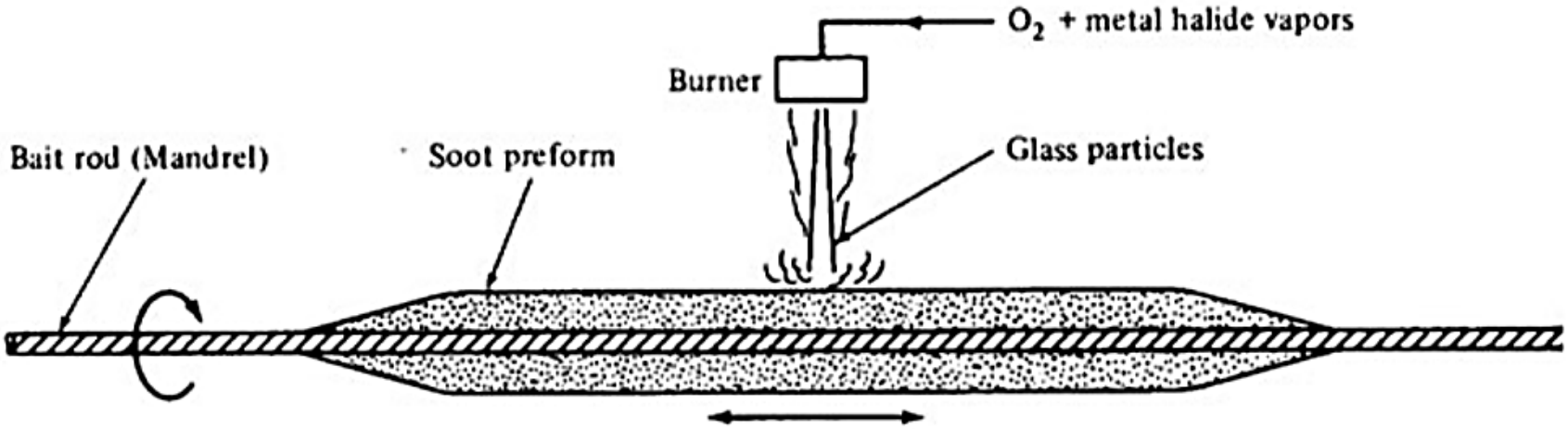
- **The preforms are typically 10-25 mm diameter**
- **60-120 cm long from which fibers are drawn**

The methods to preform are given below:

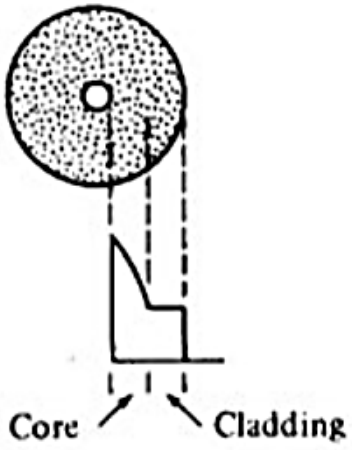
- Outside vapor phase oxidation
- Vapor phase axial deposition
- Modified chemical vapor deposition
- Plasma activated chemical vapor deposition

All these methods are based on thermal chemical vapor raction that form oxides, which are deposited as layers of glass particles called soot.

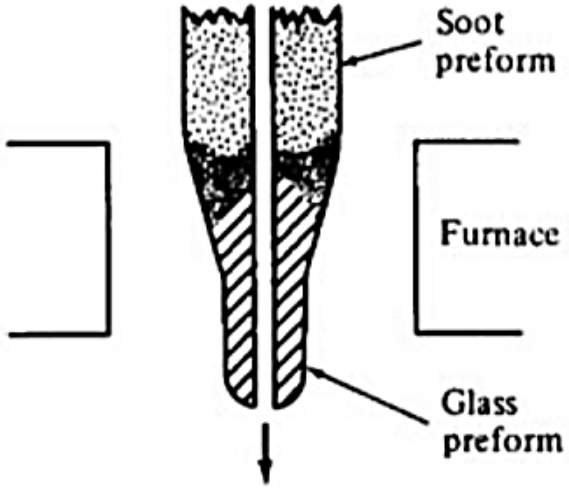
Outside Vapor Phase Oxidation



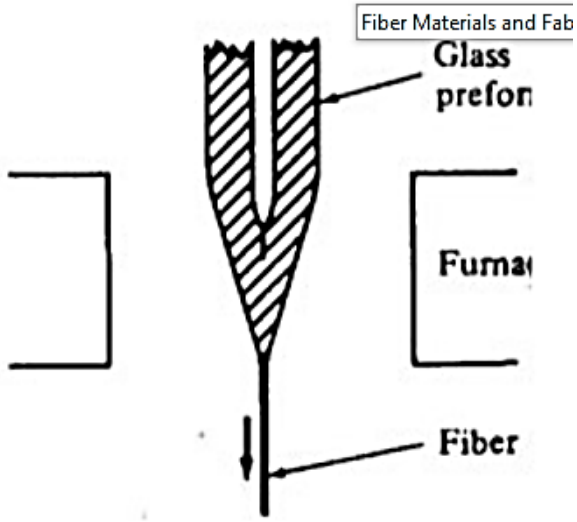
(*a*) Soot deposition



(*b*) Soot preform cross section



(*c*) Preform sintering



(*d*) Fiber drawing

Vapor Phase Axial Deposition

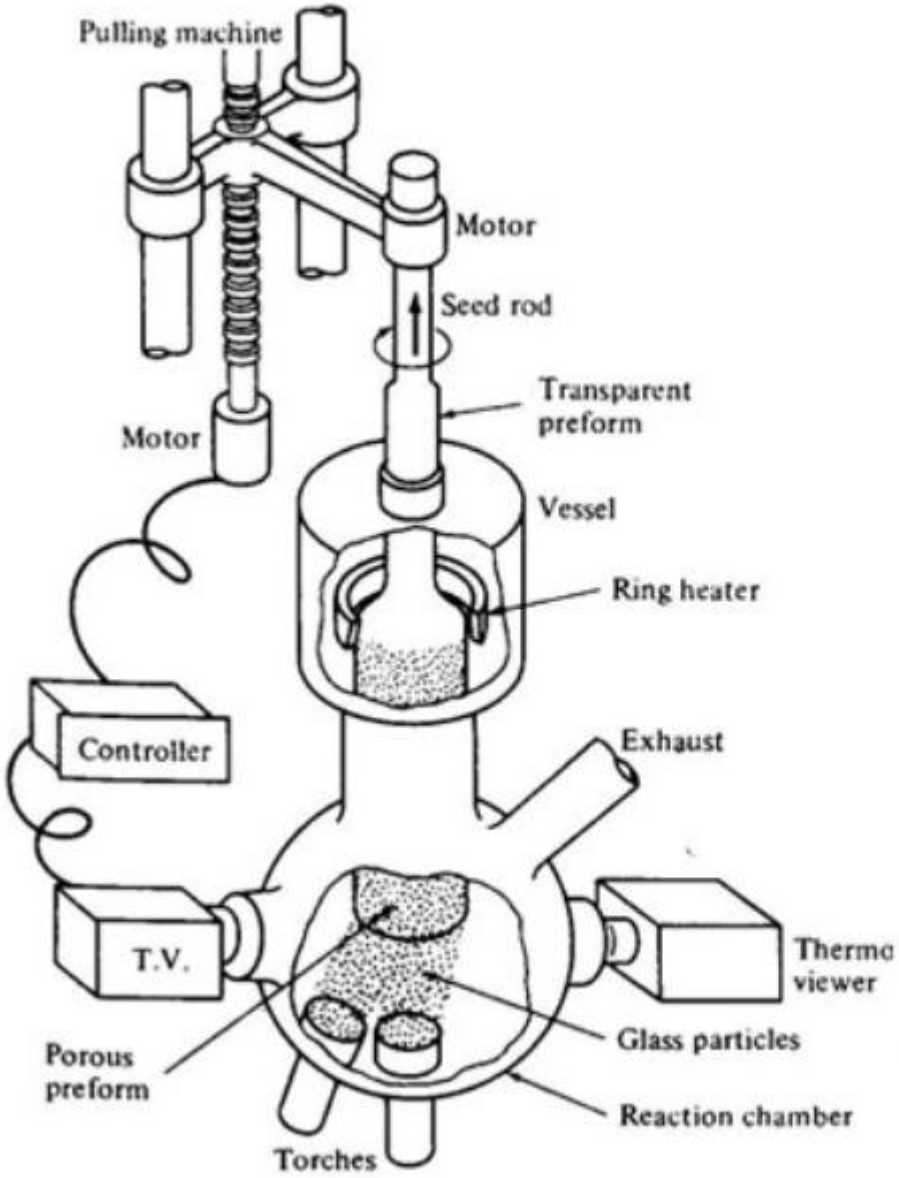
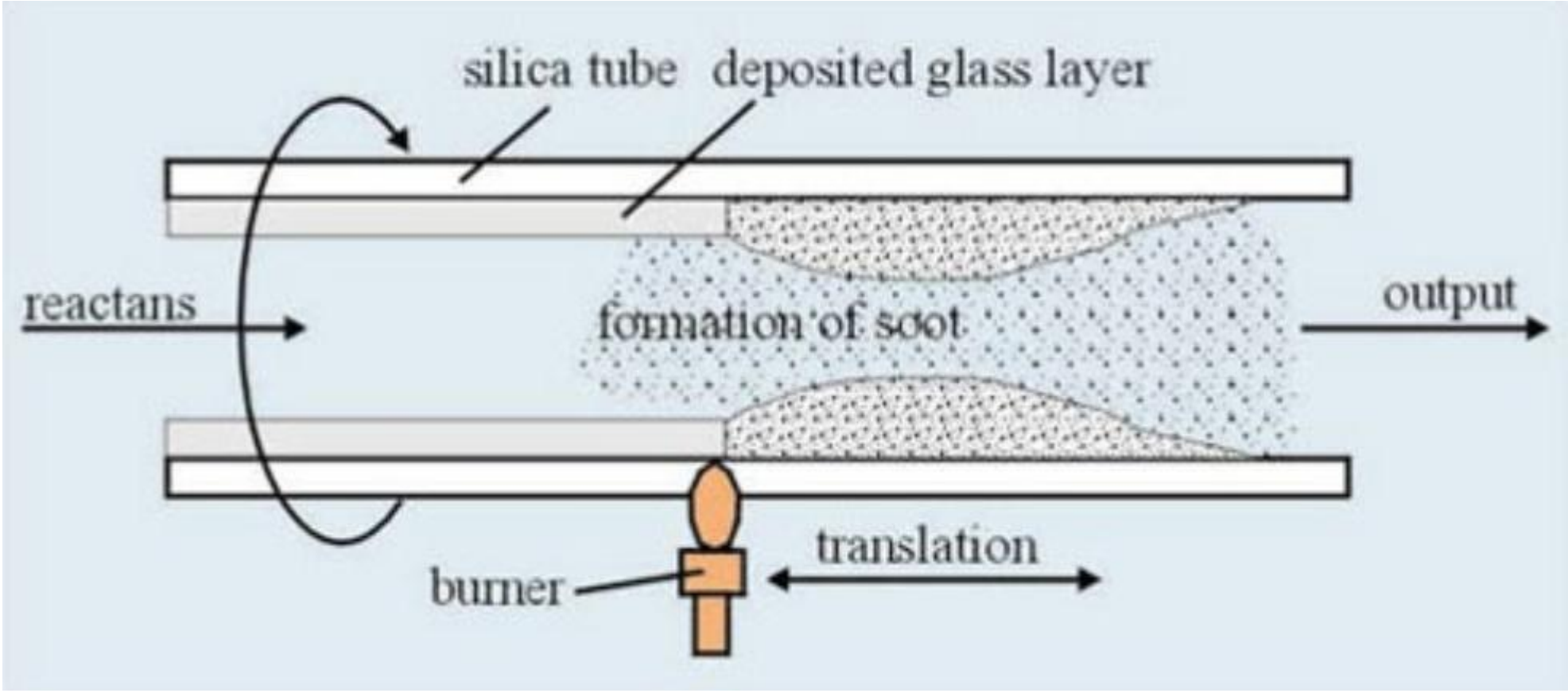


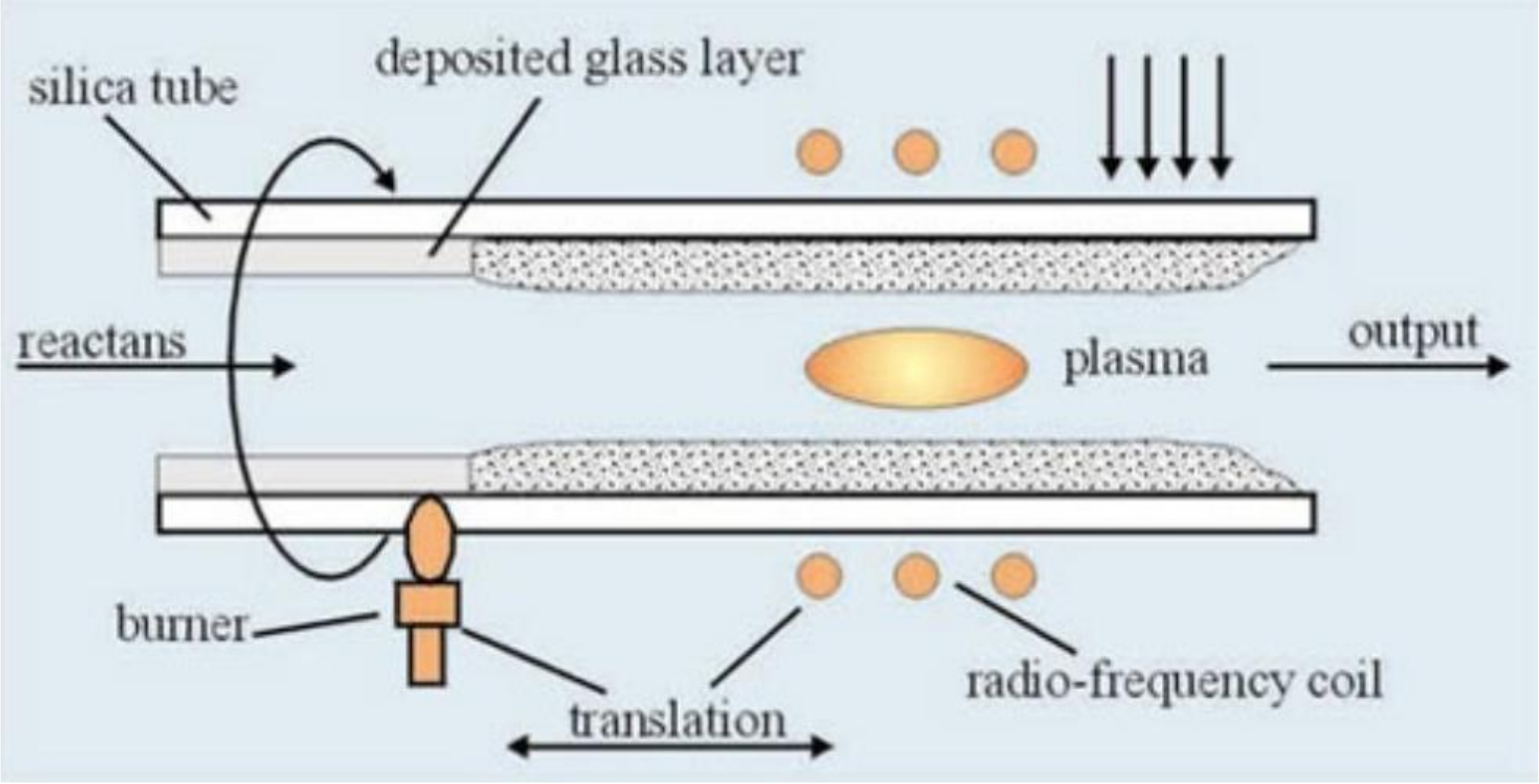
Fig. 3

Apparatus for the VAD (vapor phase axial deposition) process. (Reproduced with permission from Izawa and Inagaki,[76] © 1980, IEEE.)

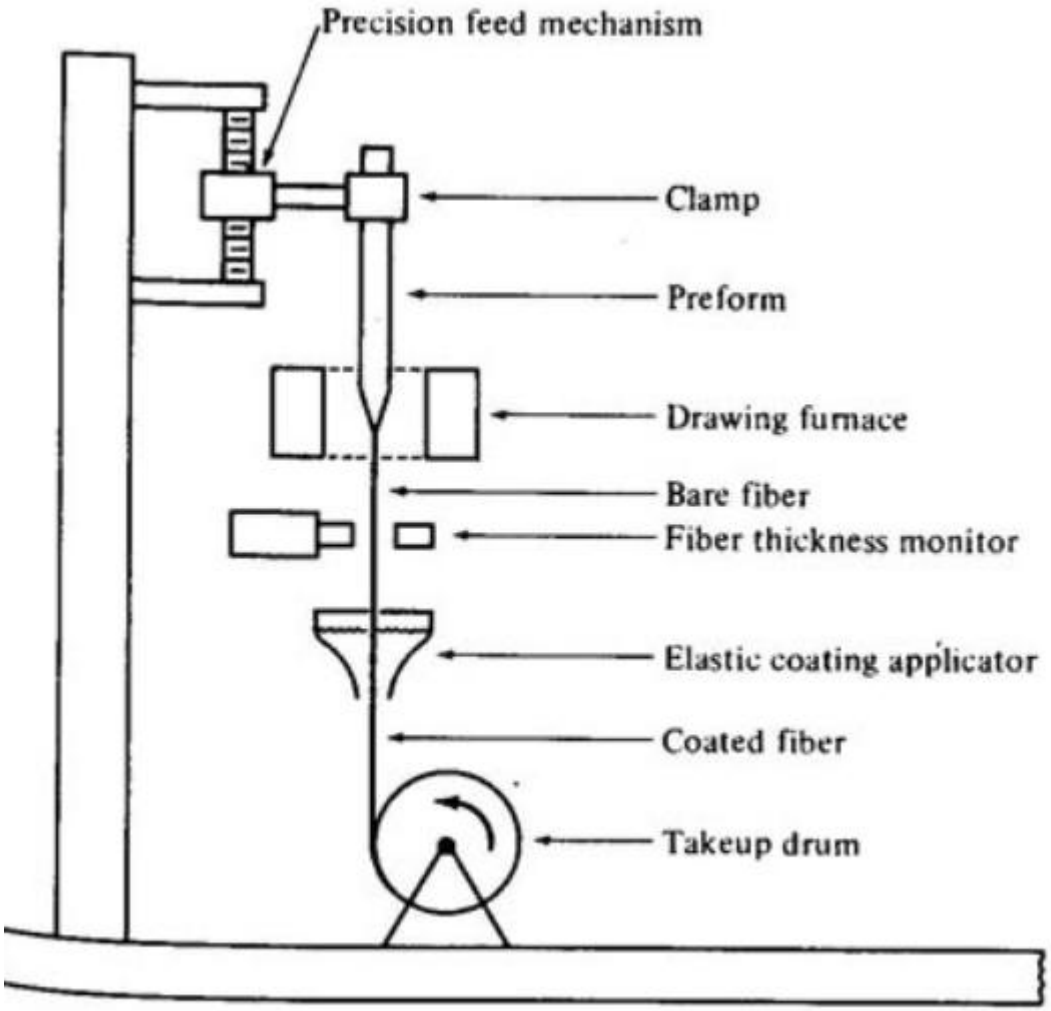
Modified chemical vapor deposition



Plasma activated chemical vapor deposition

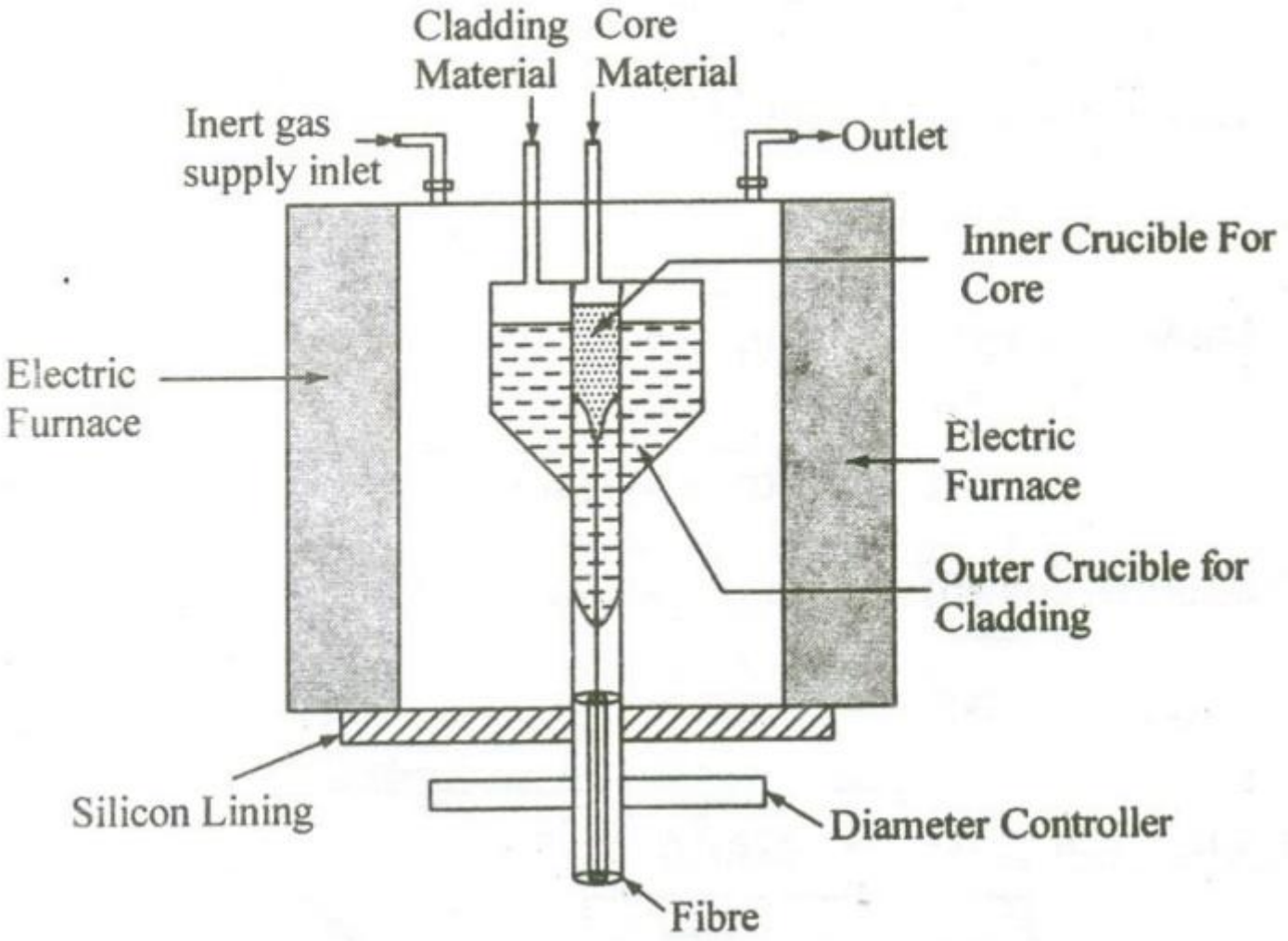


Optical Fiber Drawing



Schematic of a fiber-drawing apparatus.

Double-Crucible Method



Fiber Jacketing

